

FIRST YEAR PROJECT

Developing a Model for Diagnosis of Skin Lesions

Projects in Data Science

<https://github.com/nool-ITU/2026-PDS-Penguins>

Abstract

For the first year project, we have been presented with imagery of skin lesions from PAD-UFES-20 by ScienceDirect and corresponding masks of varying quality. This report summarises the work done and insights gained over the course of the project.

We will go over methodology for projects in data science and implement it to develop two models to classify benign and malignant lesions. The first model is based on ABC features (asymmetry, border and colour) and the second model is an extended model that adds hair coverage as a feature. We will evaluate each model using different metrics and compare the two models. Finally, we will investigate blood as a new feature and examine its impact on improving the two models.

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1 Introduction

Skin cancer is a very common type of cancer, caused by abnormal skin cells that grow uncontrollably, often as a result of sun exposure to ultraviolet (UV) radiation over a long period of time. The most common types of skin cancers are basal cell carcinoma (BCC), squamous cell carcinoma (SCC), and melanoma.[4] Melanoma is less common than other skin cancers, but it is particularly dangerous because it can spread rapidly if not detected early. Early diagnosis significantly improves treatment outcomes, so accurate assessment of lesions is an important part of dermatology.[2]

Dermatologists commonly apply the ABCD rule to assess suspicious skin lesions. The rule considers asymmetry, border irregularity, color variation, and diameter for assessment of a potential cancer of a lesion. In general, irregular shapes, uneven borders or multiple colors are considered a higher risk of cancer.[3] In recent years, Computer aided diagnosis systems for analyzing medical images have drawn considerable attention from researchers. These systems are designed to help classify skin lesions by extracting image features using machine learning techniques. Previous studies have also shown the importance of lesion segmentation and feature extraction in improving classification performance.[2]

In this project, we explore the use of image features for skin lesion classification. ABCD rule-based features such as asymmetry, border, and color variation are extracted from the clinical images and used in training the machine learning model that would be able to classify the lesions into malignant and benign lesions. These models calculate the probability of the lesion being cancerous or non-cancerous depending on its visual features. Furthermore, an extended method of classification is also developed using pre-processed images in which hair is removed. This will allow the project to evaluate whether hair removal improves feature extraction quality and overall classification performance.

2 Dataset

In this project we have used two datasets, one is the dataset for the skin lesion images, and one for annotations of pen marks and amount of hair present in the images.

2.1 Images Dataset

We used the PAD-UFES-20 [1] skin lesion dataset for this project. The dataset contains 2103

samples of six different types of skin lesions collected from 1373 unique patients. These images show skin lesions distributed across 6 different diagnoses labelled with: Basal Cell Carcinoma (BCC), Squamous Cell Carcinoma (SCC), Actinic Keratosis (ACK), Seborrheic Keratosis (SEK), Malignant Melanoma (MEL) and Melanocytic Nevus (NEV). The dataset splits into three benign diseases and three malignant cancers.

- **Cancerous:**

- Basal Cell Carcinoma (BCC)
- Squamous Cell Carcinoma (SCC)
- Malignant Melanoma (MEL)

- **Non-cancerous:**

- Actinic Keratosis (ACK)
- Seborrheic Keratosis (SEK)
- Melanocytic Nevus (NEV)

Out of 2103 total images, 1,055 are benign and 1,048 are malignant. Within the malignant group, Basal Cell Carcinoma (BCC) contains 815 images, Squamous Cell Carcinoma (SCC) has 1184, and Malignant Melanoma (MEL) has just 49.

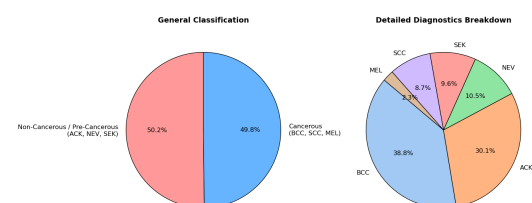


Figure 1: Diagnostic Analysis: Cancerous vs. Non-Cancerous Distribution

The quality of our images and masks varied significantly. While some masks accurately captured the exact shape of the lesion, others were oversimplified and less precise. To improve model performance, we manually reviewed our dataset to ensure proper image-to-mask alignment. We excluded poorly aligned masks, no clear lesions, or more than three lesions. Additionally, we added a new metadata column to specifically track images containing 2-3 lesions.

- useable: Accurately segmented, contains 2-3 distinguishable lesions(1865)
- deleted: Misaligned masks, no clear lesions, or more than 3 lesions(237)



Figure 2: mask (with more than 3 lesions)

2.2 Annotations dataset

We were provided with a dataset of annotations of pen marks and hair amount by 4-5 annotators. For hair amount, each annotator has given a value 0, 1, 2 and 3 to each image, where 0 means no hair, 1 means little hair, 2 means some hair, and 3 means a lot of hair is present in the image. For pen marks, each annotator has either given a value of 0 or 1, where 0 means that no pen mark are present and 1 means that a pen mark is present in the image.

To assess the degree of agreement between the annotators, we used Cohen Kappa score to calculate the degree of agreement between pairs of annotators. Since no label is given by the 5th annotator for many images, we have to consider agreement between 4 annotators and between 5 annotators but excluding the images with no 5th annotator to avoid NaN values.

The average Cohen Kappa score for 4 hair annotators is 0.742 (and 0.738 for 5 annotators), while the average score for 4 pen annotators is 0.82 (and 0.873 for 5 annotators). While the scores for both pen marks and hair annotations show a good level of agreement, the score for amount of hair is lower than the score for pen marks. There are many reasons for the score for amount of hair be lower than the score for pen marks. Big variations in the distribution of hair in some images led to different labels given by different annotators. Another reason for disagreements is that there are no clear boundaries between the labels, such that an image with barely noticeable hair can be given both label 0 or label 1 depending on the interpretation of the annotator. Pen marks on the other hand are easier to classify, yet pen marks can be confused with blood vessels or injuries, which may interpret the average Cohen Kappa score for pen marks labels.

3 Model

3.1 Feature Extraction

In order to develop a model for cancer detection from a provided image, we need to write the code

to extract the features that are commonly observed in cancerous lesions. The ABCD rule of dermoscopy is used to quantify the clinical features of skin lesions to help differentiate between benign and malignant moles. It analyzes asymmetry, border irregularity, color and dermoscopic structure. Our code tries to give a numerical value to these aspects. The values extracted from the original images, using our code, are used to train the baseline model. The extended model instead processes the original images by applying shortcuts that remove the hair that covering the lesion. By doing so, when we extract the features, we obtain better results that can help us improve the model by reducing the noise.

3.1.1 ABC Features

Feature A - Asymmetry

Asymmetry measures how similar a lesion is across different axes. A symmetric lesion appears the same when divided into two halves, whereas an asymmetric lesion is uneven and irregular in shape. In skin lesion analysis, asymmetry is clinically relevant because cancerous lesions are often more irregular and asymmetric than non-cancerous lesions, which generally tend to have smoother and more balanced shapes.

The asymmetry score is calculated using the binary lesion mask obtained after segmentation. First, the `cut_mask()` function removes empty background regions, allowing the analysis to focus primarily on the lesion. The midpoint of the cropped lesion is then determined using the `midpointGroup9()` function and used as the folding axis for symmetry comparison.

The lesion mask is separated into two halves: upper/lower and left/right. One half is flipped and compared to the other half using logical XOR operations. The XOR operation identifies pixels that do not overlap between the two halves, which correspond to asymmetric regions. The asymmetry score is calculated using the total number of non-overlapping pixels.

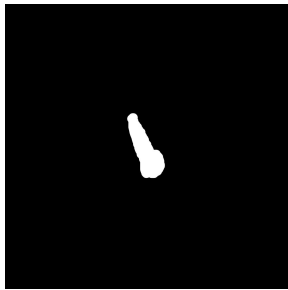
$$\text{asymmetry score} = \frac{h + v}{2 \cdot t}$$

where h = horizontal asymmetry pixels, v = vertical asymmetry pixels, and t = total pixels. A score near to zero suggests high symmetry, whereas a score close to one indicates high asymmetry. To strengthen the feature, asymmetry is calculated after rotating the lesion mask across several angles with the `rotation.asymmetry()` method. The final asymmetry feature is calculated by averaging all rotation scores using `mean_asymmetry()`. This eliminates orientation

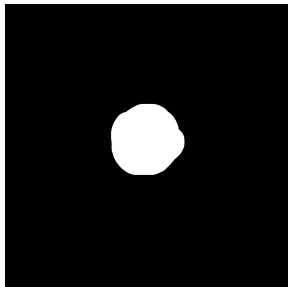
reliance and allows for more stable measurement of lesion irregularity.

The computed asymmetry scores generally align well with visual inspection of the lesions. Lesions that appear visually more symmetric tend to have lower asymmetry values, while lesions with irregular or uneven shapes tend to produce higher scores.

We can see that some of the highest asymmetry values come from non-cancerous lesions, while cancerous lesions have very low asymmetry values. This is opposite to the expected pattern, as malignant lesions normally exhibit greater asymmetry. Overall, this indicates that asymmetry alone does not consistently align with the skin cancer label.



(a) Non-cancerous: Asymmetry 0.88



(b) Cancerous: Asymmetry 0.036

Figure 3: Comparison of conflicting asymmetry scores.

One significant limitation of the asymmetry feature occurs in images with multiple unconnected lesions. Because the method examines the full binary mask as a single object, the shape analysis takes into account the gaps between individual lesions. This may affect the observed structure, making the entire region appear highly asymmetric, even if the individual lesions are rather symmetric. As a result, the computed asymmetry score may be falsely exaggerated and do not adequately reflect the true structure of the lesions. This effect is more noticeable when the image consists of many small, separated lesions.

Feature B - Border

In order to find some values that can represent how much a border can be considered irregular, circular or spiky we decided to build two function that measure compactness and convexity.

Compactness

To quantify the compactness features of the lesions, we implemented a function. First, we calculate the area by summing the pixels within the binary mask. Next, we estimate the perimeter by calculating the difference between the original mask and its eroded version, which removes the pixels along the border. The return value is then computed using the following formula:

$$\text{compactness score} = \frac{\text{perimeter}^2}{4 \cdot \pi \cdot \text{area}}$$

This formula is the reciprocal of the isoperimetric quotient that compares the area of a shape to the area of a circle with the same perimeter; since a circle has the smallest perimeter for a given area, higher values of this metric indicate that the lesion is less compact and more irregular in shape.

When testing the function on individual images, we observed that the compactness metric is not representative when multiple lesions are present. If we consider an image with multiple lesions totaling an area, the total perimeter will be significantly higher than that of an image containing a single lesion of the same area, even if each individual lesion is perfectly round. This occurs due to the isoperimetric inequality, which dictates that for a fixed total area, a single circular shape minimizes the perimeter, while multiple separate shapes result in a higher cumulative perimeter.

Convexity

To quantify the convexity features of the lesions, we implemented a function. First, we get the coordinates of all the pixel in the lesion mask, then we calculate the area by summing the pixels within the binary mask. Next, we compute the convex hull of the lesion pixels, which is the smallest convex polygon that encloses a shape, and its area. The return value is then computed using the following formula:

$$\text{convexity score} = \frac{\text{lesion area}}{\text{convex hull area}}$$

When testing the function on individual images, we observed that the compactness metric is not

representative when multiple lesions are present. If we consider an image with multiple distant lesions, the convexity hull derived by the function will also include the space that separate the lesion. In conclusion this code will not be applied on every single lesion and even if we consider a picture with 2 or 3 perfectly convex lesion, the returned value will result high compared to the one computed for concave single lesion.

Feature C - Color

Multicolor

The multicolor rate feature evaluates the amount of variation in color within a lesion. First, the lesion image is masked, allowing only lesion pixels to be evaluated. The KMeans clustering algorithm is then used to group similar colors into dominating color regions. After identifying the primary colors in the lesion, the Euclidean distance between all color cluster centers is calculated in RGB space, and the highest distance is provided as the multicolor rate. Higher values suggest greater color diversity, while lower values correspond to more uniformly colored lesions. This characteristic is important in skin lesion analysis since malignant lesions have inconsistent pigmentation and many uneven colors, whereas benign lesions have a more constant color.

Class	Mean	Standard deviation
Non-Cancerous	95.105	34.901
Cancerous	93.766	34.591

Figure 4: Distribution of classes across Multicolor feature

The results indicate relatively similar values for both classifications, with cancerous lesions having a mean multicolor score of 93.77 and non-cancerous lesions having a mean of 95.11. The standard deviations are also nearly equal, showing that both groups experience similar variability. Overall, there is no significant separation between the two groups based on the multicolor feature.

Hue Entropy

The hue entropy feature measures the diversity and randomness of the color distribution inside the lesion. The image is first transformed from RGB to HSV color space, then the hue channel is extracted. A histogram of hue values within the lesion mask is then generated, and Shannon entropy is used to determine how scattered the hue distribution is. Lesions with a variety of colors

produce greater entropy values, whereas lesions with more uniform coloring produce lower values. Hue entropy can assist capture the uneven pigmentation patterns found in malignant lesions.

Red and Blue Color Quantification

The functions `measure_red_in_lesion()` and `measure_blue_in_lesion()` are implemented to quantify the percentage of respectively red and blue color components within a given lesion. Both functions follow the same processing pipeline: first the input image is converted to HSV, then we create a mask that contain pixel in between the ranges of color we specified. Next, we applied the logic “AND” operator to select only the parts of the mask that are both on the lesion mask and in the color mask. Then return the percentage of the mask that is covered by the respective red or blue pixels.

Red and Blue Color Intensity

The two functions `get_average_red_intensity()` and `get_average_blue_intensity()` are implemented to quantify the mean of the red and blue value in RGB within a given lesion. Both functions follow the same processing pipeline: first the input image is cut by the mask to isolate the lesion, then we extract a matrix that give us the value of red (indexed as 0 in RGB) or blue (indexed as 2 in RGB). Next, we select only the values higher than 0 so that we can return the mean of the extracted values. If the mask contains no valid pixels, it returns 0.0.

HSV Variance

This function is designed to return the variance of the three values that compose the HSV color space. To work, it needs to call another function called `get_hsv_means()`. The function `get_hsv_means()` use `slic_segmentation()` to divide the image (cut by the mask) in 50 segments. Within each of them, the function computes the mean of the three values and stores them in an array. Note that the hue value uses `circmean` because its value can be seen as a chromatic circle where 1 and 0 are equivalent. The function `hsv_var()` takes the array generated by `get_hsv_means()` and calculates the variance of these values calculated across the 50 segments. It returns three values: `hue_var`, `sat_var`, and `val_var`. If the result is a high variance, it indicates that the colors detected in the image are different from each other.

3.1.2 Feature Analysis

Standardized boxplots were created for all features to analyze the distributions of the extracted features across benign and malignant lesions. Standardization was applied because all features had different numerical units. The boxplots showed significant overlap between the two classes across most features, as seen by the similar medians and interquartile ranges observed for both red and blue boxes for the majority of the features. This means that the extracted features do not provide a clear linear separation between the classes.

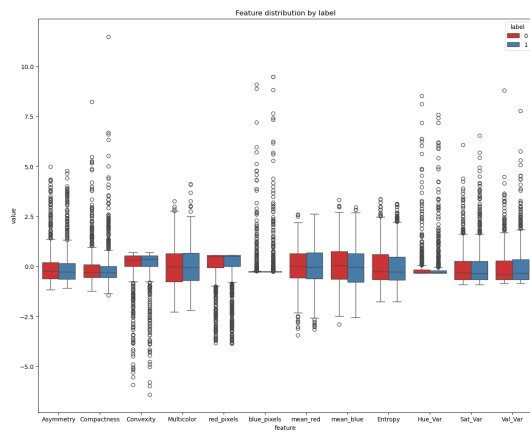


Figure 5: boxplot distribution of extracted features grouped by lesion class

Additionally, there were a lot of severe outliers and highly skewed distributions in some features such as compactness, Hue variation, and blue pixels. Such outliers may affect distance-based classifiers such as KNN and show significant variation in lesion appearance throughout the dataset.

Overall, there is a significant overlap in the feature distributions between the two classes, which could be partially caused by image noise like hair, pen marks, and lighting variations that affect the appearance of lesions and the accuracy of extracted features. Therefore, rather than depending on distinct linear separation, we need models that are susceptible to noisy and overlapping feature spaces.

Strong class overlap was shown by standardized boxplots, indicating low separability of individual features. To further investigate the shape of the features, pairwise scatter plots (look in the appendix) were created to evaluate the potential relationships between features. The scatter plots show that some feature pairs, such as Asymmetry and convexity, mean red and mean blue pairs, follow extremely similar trends across both classes. This implies that rather than representing sepa-

rate information, these features might be capturing overlapping visual characteristics of the lesions. Because of this redundancy, we need to be careful when including highly correlated feature pairs in the model, as they provide overlapping information. Using them together can add new signal while increasing the risk of overfitting and decreasing model reliability.

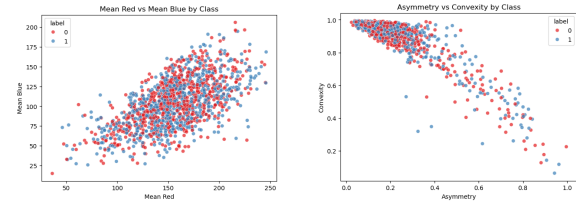


Figure 6: Scatter distributions of highly correlated features.

In order to quantify correlations over all features in a more systematic way, a correlation matrix was calculated to support the pairwise scatter plot analysis. As seen from the matrix, it confirms our previous analysis about the presence of multiple correlated feature pairs. Furthermore, we can clearly see that asymmetry and compactness, entropy and hue variation show strong positive correlation whereas compactness and convexity, red pixels and blue pixels show strong negative correlation. Overall, there is a weak correlation between individual features and the target label throughout the matrix.

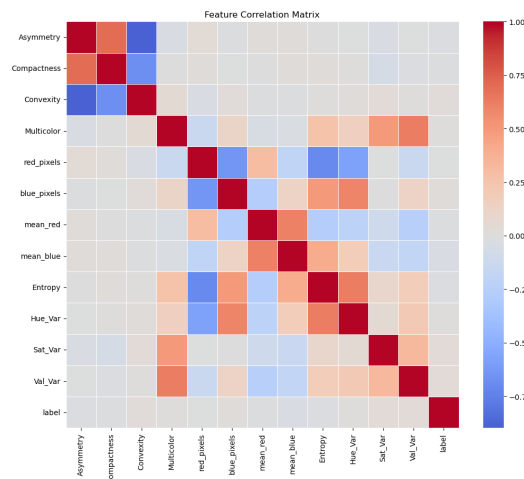


Figure 7: Correlation matrix of all features

3.2 Data Cleaning

While looking at the images present in the dataset, we noticed that some of the images were associated with a mask that did not correspond to

their actual shape. For this reason, we decided to delete the picture with mask that were either too large or too small. Another group of images that we decided to exclude from our dataset consist of those that have more than 3 lesions in it. Cutting out images with four or more lesions helps avoid model confusion. Usually in medical datasets the labels and values refer to the most critical lesion so having multiple lesions can create noise and provide wrong information to the model.

On the other hand, keeping images with two or three lesions, rather than strictly sticking to single-lesion photos, prevents us from shrinking our dataset too much. It also forces the model to be more robust, teaching it to look past minor background distractions just like a doctor would in a real clinical setting. Since we encountered issues with how the asymmetry and border features were applied to images with two or three lesions, we decided to replace their asymmetry, convexity, and compactness values with the mean of all the other values present in the training dataset.

To implement the code for this adjustment, we annotated all images by adding a Boolean column named `2_3_lesion`. We chose not to delete these images because their remaining data is still highly valuable for training the model on color features, and doing so allowed us to avoid losing too much data. Using the mean provides a neutral baseline that prevents the algorithm from crashing due to missing values, without distorting the model with extreme or misleading data like zeros. This approach successfully neutralizes the problematic geometric features while keeping the dataset size intact and preserving the valuable color information.

3.3 Baseline Model

To find the best baseline model for our dataset, we have evaluated the following machine learning models:

- k-nearest neighbors
- Logistic Regression
- Random forest

The dataset was initially divided into five splits using a stratified GroupKFold method based on the variable's 'diagnosis' and '2_3_lesions'. This ensures to have the same proportion of benign and malignant cases, as well as a balanced distribution of 2-lesion and 3-lesion images in all the splits. The splits were labeled 'a', 'b', 'c', 'd', and 'e'. To prevent data leakage, all photos from the same patient were allocated to the same split, which meant that a patient

may appear in either the training set (splits a–d) or the test set (split e), but not both.

Since these splits were already predefined, our model training used only four training splits (a–d) during cross-validation. Therefore, during hyperparameter tuning, we used `stratifiedGroupKFold` with four folds, which corresponded exactly to the four available training groups. The final evaluation was performed on the completely unseen test split (e).

We have implemented a single machine learning workflow that ensures to have a consistent and valid evaluation of all models. Each model was integrated into a pipeline which contains a `StandardScaler` for normalization of features, which is extremely important for classifiers like k-NN and logistic regression, `Sequential feature Selection (SFS)` for selecting the most informative combination of features, The pipeline was then fitted on the training data. For hyperparameter tuning, `GridSearchCV` combined with `GroupKFold` was used to systematically evaluate different combinations of hyperparameters and return the best combination. The F1-score was used as the optimization metric to balance false positives and false negatives. After selecting the best hyperparameters for each model, we retrained the model on the full training data (splits a–d) and calculated f1-score. The final evaluation was performed on the test data (e). The evaluation was analysed based on AUC, F1-score, precision, recall, and confusion matrices with a focus on predicted probabilities and not hard labels.

Model	Accuracy	AUC	Recall	Precision	F1-score
Random Forest	0.5121	0.5148	0.6667	0.5263	0.5882
kNN	0.5282	0.5239	0.6154	0.5430	0.5769
Logistic Regression	0.5228	0.5016	0.9385	0.5244	0.6728

Figure 8: Performance metric comparison of the baseline models.

K-Nearest Neighbors

We evaluated the kNN classifier in order to build our baseline model. This model classifies a lesion based on the label of its closest neighbors in the space created by features. For this reason, it is very sensitive to how well the classes are separated. The statistics that we obtained for this model once trained on the training data using cross validation and tested with the test data, are too close to those obtained with random performance. This suggested that the handcrafted features do not form well-separated clusters for benign and malignant lesions.

Logistic Regression

Logistic Regression was included as a baseline linear classifier capable of producing probability

estimates. We trained the model using cross validation, but the model is still not able to predict the class of a lesion. From our low value for AUC and Accuracy we can clearly say that the features do not provide enough linear signal to distinguish between benign and malignant lesions. This indicated that the relationship between the features and the target is not linearly separable.

Random Forest

Random Forest was tested to capture non-linear relationships and interactions between features. It is robust to noise and provides feature importance scores, making it a strong candidate for analyzing features data that is not clearly separated for cancerous and benign lesions. We used cross validation in order to build the model with the best hyperparameters.

We chose kNN as the final model because its AUC and Accuracy have the highest values even though they are still close to randomness, but this was the best model we could find given the 12 features we extracted.

3.4 Extended Model

The extended classification method is exactly the same machine-learning workflow as the baseline approach, using the same pipelines, classifiers (k-Nearest Neighbors, Logistic Regression, and Random Forest), and the same GroupKFold validation which provides a fair and controlled comparison. The same preprocessing steps (StandardScaler, Sequential Feature Selection, and GridSearchCV optimized for the F1-score) were applied to all models, and the final evaluation was again performed on the unseen test data (e) using predicted probabilities.

The additions that were made in the extended method are that all features were again extracted from preprocessed images (hair removed images) to reduce noise and an additional feature, named hair feature which represents the hair coverage value of each image, was included to quantify the amount of hair in the original image.

3.4.1 Hair Segmentation & Removal of Pen Marks

The hair in the images was segmented using BlackHat and TopHat operations with enhanced contrast. BlackHat is generally used for detecting black colour, whilst TopHat is generally used for detecting white colour. These were used to remove black hair and white hair from the images. The two methods use the two morphological operations erosion and dilation. The main parameters of the two methods are the kernel, threshold,

and radius. However, the Black/TopHat methods alone seemed not to perform as expected due to low image quality.

To enhance the performance of BlackHat and TopHat, the contrast of the images was enhanced in order to accentuate the black and white hairs before applying BlackHat and TopHat. The enhancement of the contrast was done using a SciKit Image function named exposure.[5] However, enhancing the contrast also led to the highlighting of other parts of the skin texture, especially in images that were taken under bad lighting conditions. This means that BlackHat and TopHat were also removing some of the image texture, and hence reducing the quality of the image by blurring it in many cases.

To develop an automatic method to segment hair from masks and extract hair coverage as a feature, we calculated the hair coverage in each image and compared it to the mean of the annotations of the hair amount labels in the images that we used as the ground truth. The hair coverage values and annotations' labels are visualised by the plot below:

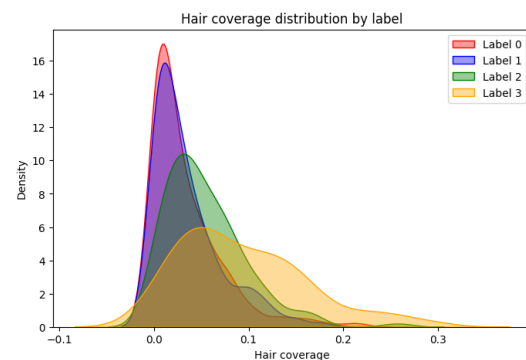


Figure 9: Hair coverage plot

We set the final parameters based on visual investigation of the images before and after processing and using the plot above. From the plot, it can be seen that the curves for images with a hair amount of 0 or 1 significantly overlap and can hardly be separated. A reason for this might be because the difference between images with zero or one images is small. To ensure that the automatic hair segmentation function removes more hair from images with a lot of hair and little to no hair in images with little to no hair. we set different parameters for the kernel and threshold depending on the hair coverage value of an image. We set lower values for the kernel and higher values for the threshold for images with low hair coverage. We increased the kernel and decreased the threshold for higher hair coverage values. For images with hair coverage higher than 0.18, we

did not segment hair, as it only blurs the image without segmenting hair.

For removing pen marks, we relied on the same functions used for hair segmentation for several reasons. We tried to BlackHat and with the red channel of the images to enhance the visibility of the blue pen marks. However, this did not completely eliminate the pen marks but left a visible trace of blue marks after removal. Meanwhile, the function was significantly blurring and destroying the texture of the skin lesions, especially when there was little to no pen marks. In images, the function was detecting more than half of the image to be pen marks. The function seemed to be too volatile to be used for pen marks removal. While the functions used for hair segmentation failed to remove pen marks in many cases, they were successful in removing them in other images.

After implementing the hair segmentation method, it seemed that it failed to remove hair, especially faint hair, in many images. It also was blurring the images or distorting the texture of the skin lesions significantly. Meanwhile, it was performing better on some other images.

3.4.2 Extended Model Results

After removing the hair from the images, we applied the described classification workflow to all three models. The table below summarizes the performance metrics obtained for each classifier on the extended features data.

Model	Accuracy	AUC	Recall	Precision	F1-score
Random Forest	0.4906	0.4841	0.6410	0.5102	0.5682
kNN	0.5040	0.5304	0.5692	0.5236	0.5455
Logistic Regression	0.5255	0.5057	0.9385	0.5259	0.6740

Figure 10: Performance metric comparison of the extended models.

Across all three classifiers, the extended model did not show significant improvement in performance. However, kNN model resulted in slightly improved metrics, yet remained close to random performance. On the other hand, both Logistic Regression and Random Forest showed no noticeable change and there was a slight decrease in AUC and F1-score. As a result, the extended method and additional features have had minimal impact on the separation of benign and malignant lesions. We decided to use kNN as our final classifier because it demonstrated some improvement, although it is slight.

4 Discussion

When comparing the confusion matrices, both models show very similar misclassification patterns. The baseline model misclassified 101 benign and 75 malignant cases, while the extended model misclassified the same number of benign cases and slightly more malignant cases (84). This shows that the additional preprocessing and Hair Coverage feature did not improve class separation.

Several reasons contributed to the little improvement of the extended model. The dataset has a significant amount of variation in lesion appearance, with many benign and malignant lesions sharing similar asymmetry, border, and color characteristics, and in some cases, benign lesions showing higher asymmetry values than malignant ones. Moreover, segmentation quality was not fully consistent across the data set, as some masks were focusing on the entire lesion and others were capturing only the center wound or damaged region, which could affect the quality of the extracted features. Furthermore, pen marks and other artifacts remained in many images, which could have caused additional noise into extracted features. Although the dataset assumes five annotators, these annotations were created by multiple people rather than the same individuals, which shows that there is some level of subjectivity and inconsistency in annotations, which were used in the process of computing hair coverage values. These issues likely made it difficult for the baseline and extended models to identify clear distinctions between benign and malignant cancers, especially when combined with the large overlap found between the majority of features.

5 Open Question

“Can detection of bleeding enhance the model performance?”

After investigating the pictures, we noticed that some images have traces of blood, which may be caused by rubbing them due to itchiness and the nature of the skin disease. This motivated us to investigate whether the presence of blood on skin lesions can be used as a feature of to separate benign from malignant lesions. The pipeline of developing a method to detect blood in images was mainly focused on two points:

1. Blood is known to be red. However, many images do not have fresh blood but dried blood, which has a brownish red colour. Therefore, the blood detector function should be able to detect both fresh and dry blood. For detecting red to brownish red colour, we used a hybrid of both

RGB and HSV colour models.

2. The blood detector should not detect random red pixels. Instead, it should detect spots or blobs of reddish pixels that could resemble blood. For detecting clusters of red pixels, we used an OpenCV function named `connectedComponentsWithStats` which is designed to which detects groups of connected pixels.

To test the quality of the blood detector function. We used the column bleed in the metadata as the ground truth to test the validity of the function. The function returns two features, which are the label of blood presence and the number of blood pixels detected. Before using the function for the baseline and extended models, we tested its quality using different metrics as shown below:

- Accuracy: 70.38%
- Precision: 48.50%
- Recall: 51.31%
- F1-score: 49.86%
- Specificity: 78.05%

While the accuracy of the blood detector function is 70 percent, we can see that the other metrics have a relatively low percentage. Because of class imbalance such that 70 percent of the skin lesions have blood, accuracy is not a reliable metric. The low percentage for precision, recall and f1 metrics is due to several reasons. One of the issues with the function is that it cannot distinguish dried blood from brownish-reddish lesions, as they both have similar colours. In this case, the function should be able to detect the texture of the red regions, as dry blood is crusty while skin lesions are usually soft. Another issue we encountered while developing the function is that some images did not seem to have blood by visually inspecting them, and the function could not detect any blood pixels, but it was labeled True in the bleed column in the metadata, which makes as suspect whether the labels in the bleed column are actually correct or not.

while developing the blood detector function, we noticed a trade-off between false negatives and false positives. For instance, increasing the minimum number of pixels led to rejecting many positives despite detecting blood pixel blobs in them. Meanwhile, decreasing the minimum limit led to increasing false positives. This trade-off cannot be handled by detecting colour and compact regions alone.

After developing the blood detector function and using it as a feature in both the baseline and

extended models, we obtained the following results:

Model	Accuracy	AUC	Recall	Precision	F1-score
Random Forest	0.4960	0.5042	0.6359	0.5145	0.5688
kNN	0.5147	0.4962	0.5487	0.5350	0.5418
Logistic Regression	0.5228	0.5016	0.9385	0.5244	0.6728

Figure 11: Performance metric comparison of the baseline models with the blood feature.

From the table above, we can see that random forest is the best model among the three with the highest AUC of 0.5042, which is lower than the original final baseline model by two percent.

Model	Accuracy	AUC	Recall	Precision	F1-score
Random Forest	0.4933	0.4956	0.6154	0.5128	0.5594
kNN	0.5040	0.5304	0.5692	0.5236	0.5455
Logistic Regression	0.5255	0.5057	0.9385	0.5259	0.6740

Figure 12: Performance metric comparison of the extended models with the blood feature.

From the table above, we can see that kNN is the best model among the three with the highest AUC of 0.5304, which is similar to the AUC of the original final extended model.

The new blood feature has not improved neither of the original final models. Since the performance of the models is low, we cannot conclude the effect of the presence of blood on classifying benign and malignant skin lesions.

6 Conclusion

In this report, we developed two models, baseline and extended, to classify skin lesions as benign or malignant. In the baseline model, we extracted ABC features (asymmetry, border, and colour) from skin lesion images. Then, we classified the images into cancerous and non-cancerous lesions using three machine learning models; k-nearest neighbors (kNN), logistic regression, and random forest. After Comparing the three classification models, the kNN model was the best with the highest AUC among the three models. In the extended model, we segmented hair from the images, and extracted hair coverage as a feature and added it to the ABC features. For the classification model, kNN again was the best model with the highest AUC and F1 scores compared to logistic regression and random forest.

However, the AUC of both the baseline and extended models was close to randomness, such that the baseline model AUC was 0.52 and the extended model AUC was 0.53. These low AUC scores were possibly due to low image quality, inconsistent masks, and unreliable annotations that

could not lead to better performance in the extended model.

We investigated the presence of blood on skin lesions as our open question, and added it as a new feature to both the baseline and the extended models. However, the results did not show a significant improvement and the AUC remained close to 0.5 in both models.

7 References

References

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8 Appendix

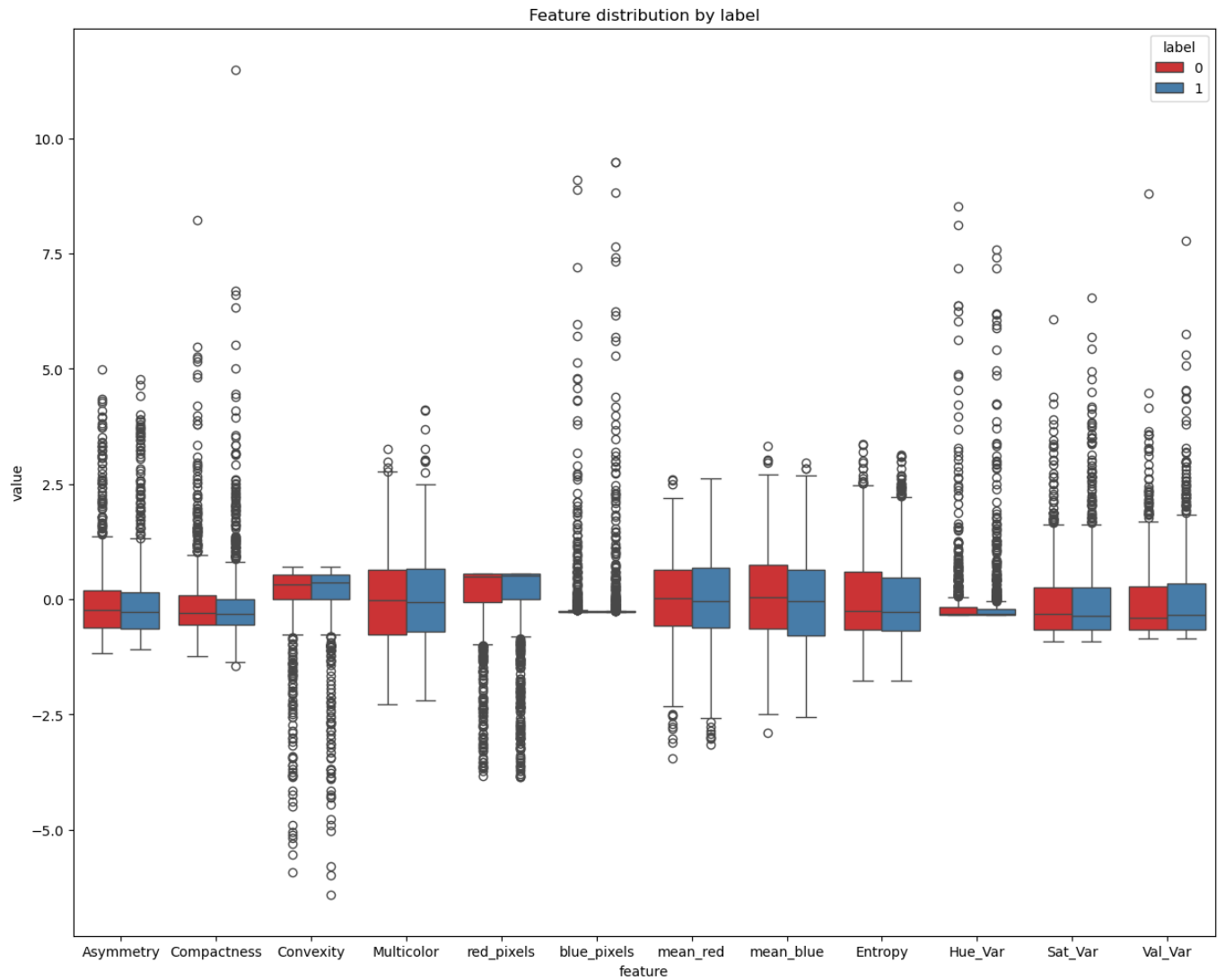


Figure 13: boxplot distribution of extracted features grouped by lesion class

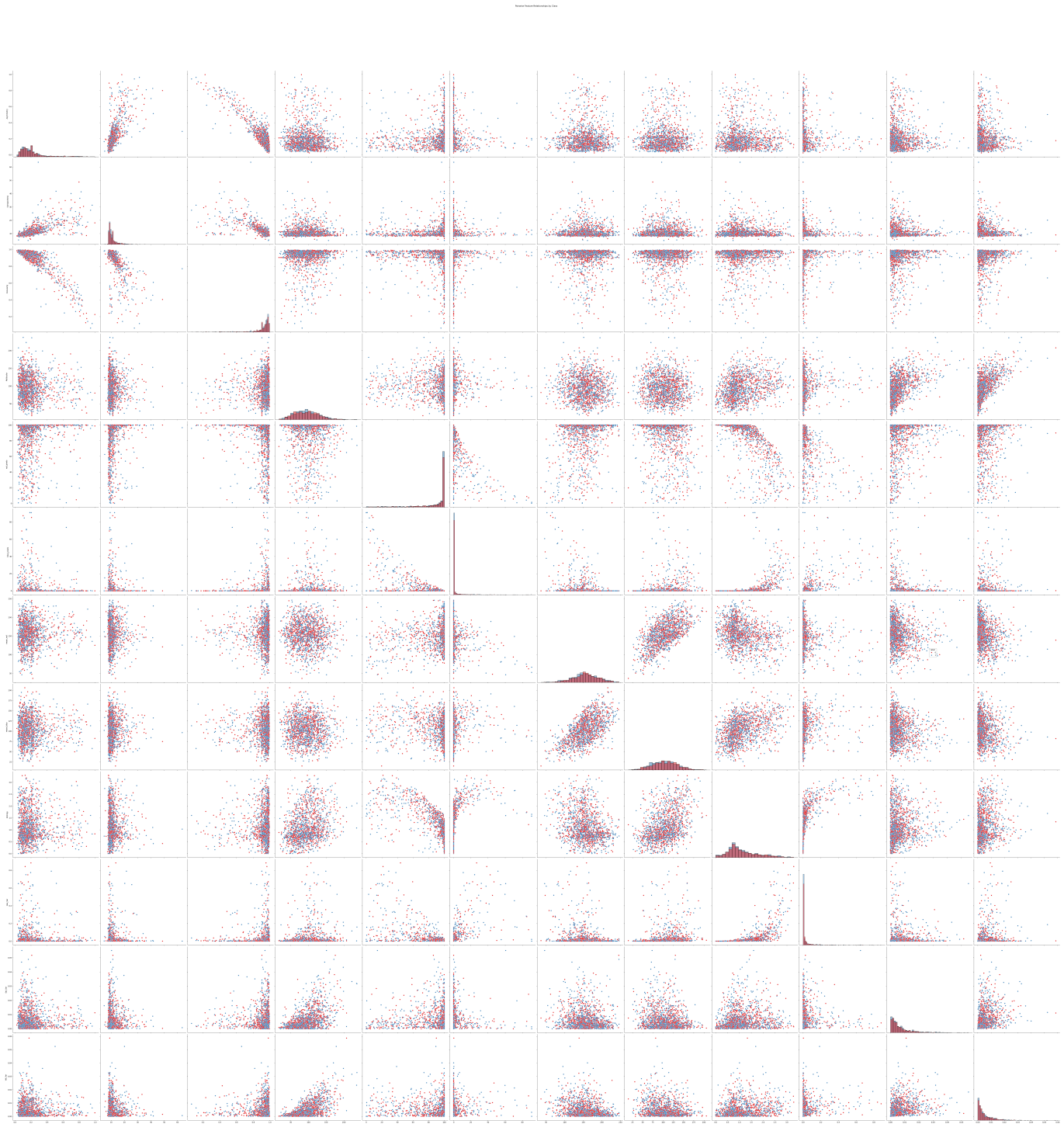


Figure 14: pairplot of all features

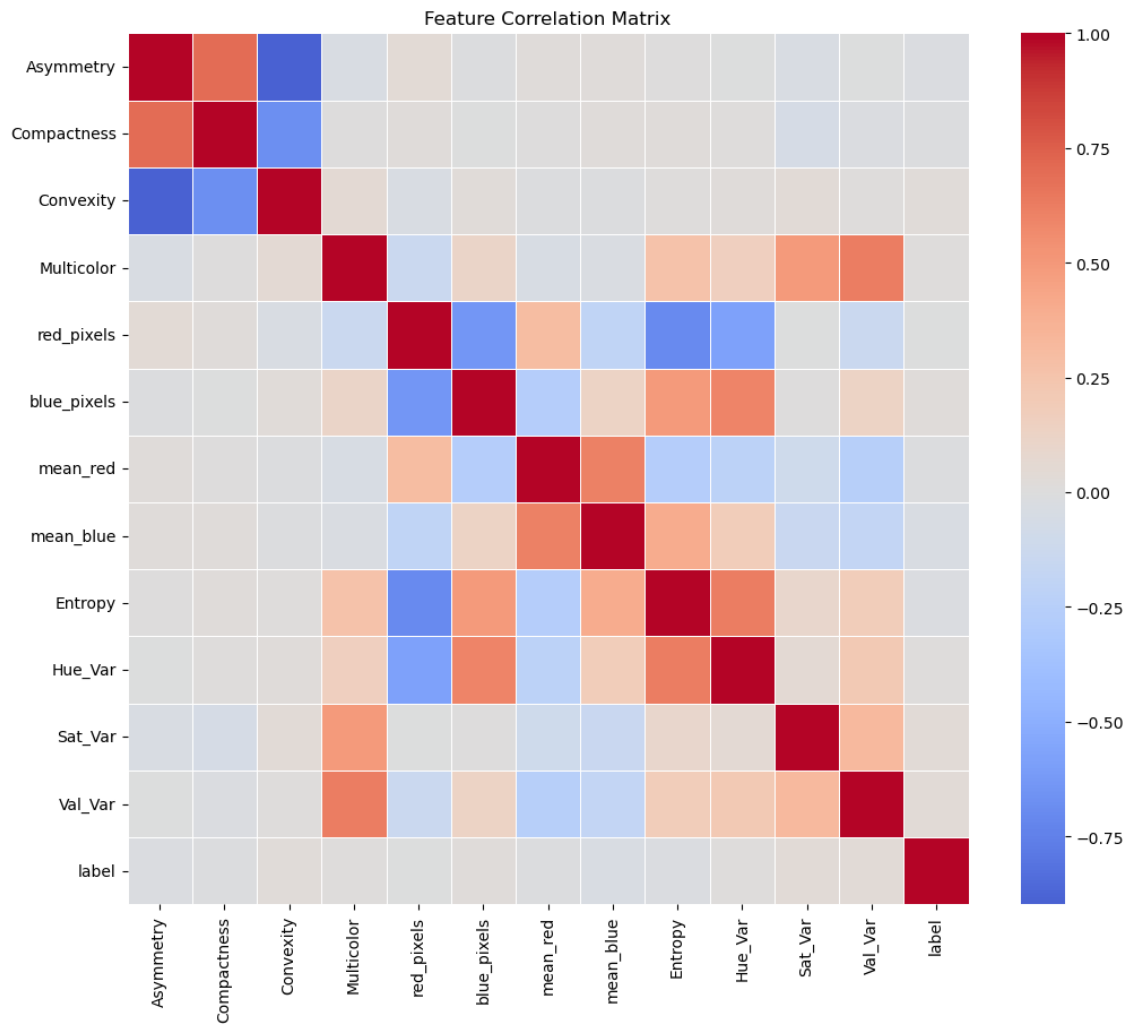


Figure 15: Correlation matrix of all features

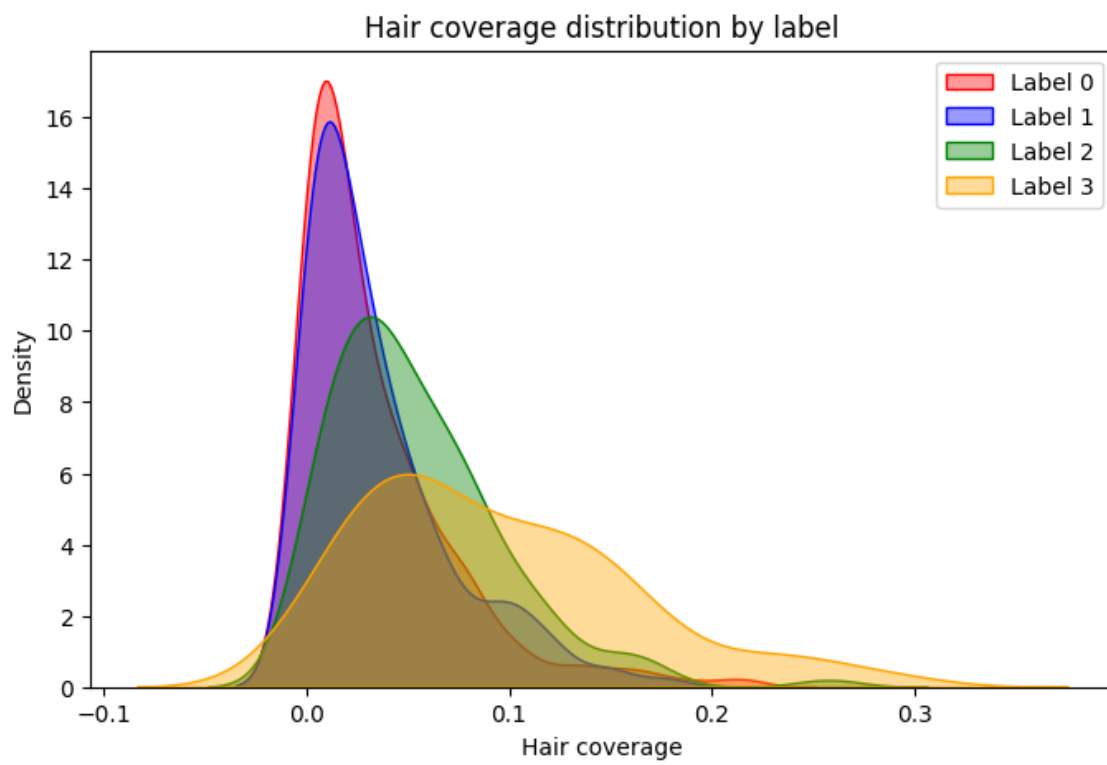


Figure 16: Distribution of hair coverage across different labels